

CO2-AT2 Assignment

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QUESTIONS

Q1 
Grocery Shopping Cart
Management System
10 marks

Q2 

Q1 **Grocery Shopping Cart Management System**

10 marks

Add products using add()
Remove purchased products using remove()
Check whether a product exists using contains() Check how many products there in the list

Online Shopping Coupon
Verification
10 marks

Q3 
ATM PIN Security Validator
10 marks

3/3 answered

Your Code

```
1 import java.util.ArrayList;
2 public class GroceryCart {
3     public static void main(String[] args) {
4         ArrayList<String> cart = new ArrayList<>();
5         cart.add("Milk");
6         cart.add("Bread");
7         cart.add("Eggs");
8         cart.add("Butter");
9         System.out.println("Cart after adding products: ");
10        System.out.println(cart);
11        String checkItem = "Milk";
12        if (cart.contains(checkItem)) {
13            System.out.println(checkItem + " is present in the cart.");
14        } else {
15            System.out.println(checkItem + " is not present in the cart.");
16        }
17        String checkItem2 = "Cheese";
18        if (cart.contains(checkItem2)) {
19            System.out.println(checkItem2 + " is present in the cart.");
20        } else {
21            System.out.println(checkItem2 + " is not present in the cart.");
22        }
23        System.out.println("Total number of products in the cart: " + cart.size());
24    }
25 }
```

Input

stdin input

Output

```
Cart after adding products:
[Milk, Bread, Eggs, Butter]
Milk is present in the cart.
Cheese is not present in the cart.
Total number of products in the
cart: 4
```

QUESTIONS

Q1Grocery Shopping Cart
Management System
10 marks**Q2**Online Shopping Coupon
Verification
10 marks**Q3**ATM PIN Security Validator
10 marks

3/3 answered

Q2 Online Shopping Coupon Verification

10 marks

An e-commerce website offers the coupon code SAVE50. Write a Java program that:

Removes extra spaces using trim()

Converts the entered code to uppercase using toUpperCase()

Compares it with the original coupon using equalsIgnoreCase()

Your Code

```
1 import java.util.*;
2 public class CouponVerification {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String coupon = "SAVE50";
6         String code = sc.nextLine();
7         code = code.trim();
8         code = code.toUpperCase();
9         if (code.equalsIgnoreCase(coupon))
10             System.out.println("Coupon Applied Successfully!");
11         else
12             System.out.println("Invalid Coupon Code!");
13     }
14 }
```

Input

SAVE50

Output

Coupon Applied Successfully!



Run



Save



QUESTIONS

Q1

Grocery Shopping Cart
Management System

10 marks

Q2

Online Shopping Coupon
Verification

10 marks

Q3

ATM PIN Security Validator

10 marks

3/3 answered

Q3 ATM PIN Security Validator

10 marks

A nationalized bank receives ATM PINs from customers through a kiosk. Develop a Java program that accepts the PIN as a string and performs the following operations:

Remove unwanted spaces using trim()

Validate that the PIN contains exactly 4 digits using matches()

Display the PIN length using length()

Your Code

```
1 import java.util.*;
2 public class ATMPinValidator {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String pin = sc.nextLine();
6         pin = pin.trim();
7         System.out.println("PIN Length: " + pin.length());
8         if (pin.matches("\\d{4}"))
9             System.out.println("Valid PIN");
10        else
11            System.out.println("Invalid PIN");
12    }
13 }
```

Input

7412

Output

PIN Length: 4
Valid PIN

